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ADVISORY CIRCULAR

SUBJECT: MANNED POWERED-LIFT (VTOL) OPERATIONS	DATE: 2026-04-16	AC NUMBER: 100-01	VERSION: 1.0

NOTE: THIS ADVISORY CIRCULAR IS PUBLISHED TO PROVIDE REGULATORY INFORMATION AND DESCRIBE ACCEPTABLE MEANS OF COMPLIANCE WITH THE GENERAL AUTHORITY OF CIVIL AVIATION REGULATIONS (GACAR).

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CHAPTER 1. INTRODUCTION

1.1 Purpose of This Advisory Circular (AC).

1.1.1 This AC describes how powered-lift will comply with specific regulations in GACAR Parts 43, 91, and 135. This AC does not provide compliance options for every regulation addressed in GACAR Part 100, as several do not require additional information. Instead, this AC provides clarification for specified regulations within GACAR Part 100, to ensure powered-lift operate safely.

1.1.2 *Powered-Lift and Applicable Aircraft Rules.* GACAR Parts 43, 91, and 135 contain some provisions that are applicable to all aircraft and some provisions that specify applicability to a particular kind of aircraft (e.g., airplane or rotorcraft). Powered-lift are aircraft as defined in GACAR § 1.1 and, as explained in GACAR § 100.301, must comply with regulations that are applicable to aircraft.

Note:

GACAR Part 100 is applicable to powered-lift aircraft and other innovative vertical takeoff and landing (VTOL) capable aircraft. Powered-lift and innovative VTOL capable aircraft are differentiated from helicopters and other traditional rotorcraft by the number of rotors or the presence of a wing-borne flight mode. All references to powered-lift in GACAR Part 100 also apply to other innovative VTOL capable aircraft unless otherwise specified.

Additionally, GACAR Part 100 is limited in its application to manned powered-lift aircraft with a maximum takeoff mass of 5,700 kg or less, and with a passenger seating configuration, excluding pilot seats, of 9 seats or less. These aircraft are referred to as "small" manned powered-lift (VTOL) aircraft and will be type certificated under GACAR Part 21 to airworthiness criteria established for "special class" aircraft as prescribed in GACAR § 21.41(b). These aircraft are neither transport category nor normal category, rather they are "special class" aircraft.

GACAR Part 100 requires powered-lift to comply with regulations that specify applicability to a particular kind of aircraft. This AC describes how powered-lift comply with specific airplane, rotorcraft, and helicopter rules contained in GACAR Parts 43, 91, and 135. Because this AC discusses the applicability of existing rules to powered-lift operations, it contains numerous references to other FAA ACs, FAA Orders, and guidance material. These documents can be found on the FAA Dynamic Regulatory System (DRS) located at https://drs.faa.gov.

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1.1.3 *Effects of Guidance*. This AC is intended to support powered-lift operations. It is not mandatory and does not constitute a regulation. The contents of this document do not have the force and effect of law and are not meant to bind the public in any way, and the document is intended only to provide information to the public regarding existing requirements under the law or agency policies.

1.1.4 *Regulatory Requirements*. The content of this AC does not change or create any additional regulatory requirements, nor does it authorize changes in, or permit deviations from, existing regulatory requirements.

1.2 Audience.
The primary audience for this AC includes pilots and operators intending to operate powered-lift in accordance with GACAR Parts 43, 91, and/or 135.

1.3 Reserved.

1.4 Background.

1.4.1 Concurrently with publication of this AC, the GACA has published GACAR Part 100 establishing the requirements for pilot certification and operation of powered-lift. Powered-lift are defined in GACAR Part 1 as heavier-than-air aircraft capable of vertical takeoff, vertical landing, and low-speed flight that depend principally on engine-driven lift devices or engine thrust for lift during these flight regimes and on nonrotating airfoil(s) for lift during horizontal flight. Powered-lift may be capable of vertical takeoff and landing (VTOL) operations or conventional takeoffs and landings, while being able to fly like an airplane during cruise flight. GACAR Part 100 establishes a regulatory structure that leverages the existing rules, removes operational barriers, and mitigates safety risks for powered-lift.

1.4.2 The FAA and GACA conducted a review of FAA ACs that may be relevant to powered-lift. These are listed in Appendix B, Related Reading Material.

1.5 Powered-Lift Flight Modes.
Powered-lift are hybrid aircraft capable of operating in a manner similar to airplanes (wing-borne flight mode) and rotorcraft (vertical-lift flight mode). The GACA refers to these modes as wing-borne flight mode and vertical-lift flight mode.

1.5.1 *Vertical-Lift Flight Mode*. Vertical-lift flight mode means a mode of flight where a powered-lift:

- Is in a configuration that allows vertical takeoff, vertical landing, and low-speed flight; and
- Depends principally on engine-driven lift devices or engine thrust for lift.

1.5.2 *Wing-Borne Flight Mode*. Wing-borne flight mode means a mode of flight where a powered-lift is not operating in the vertical-lift flight mode as defined and depends exclusively or partially on nonrotating airfoil(s) for lift during takeoff, landing, or horizontal flight.

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1.5.3 *Autorotation.* Autorotation means a rotorcraft or powered-lift flight condition in which the lifting rotor is driven entirely by action of the air when the rotorcraft or powered-lift is in motion.

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CHAPTER 2. PART 43 MAINTENANCE

2.1 Applicability of GACAR Part 43 to Powered-Lift. Powered-lift meets the definition of an aircraft as defined in GACAR § 1.1 and must comply with all GACAR Part 43 maintenance requirements that are applicable to aircraft in accordance with GACAR § 43.1, unless otherwise specified in the GACAR.

2.2 Maintenance and Inspection Provisions Applicable to Powered-Lift. The following maintenance provisions under GACAR Part 43 that pertain to rotorcraft also apply to powered-lift:

1. GACAR § 43.5(h) applies to certificate holders (CH) operating powered-lift under GACAR Part 135 in a remote area; and

2. In lieu of complying GACAR § 43.23(b), each person performing an inspection required by GACAR Part 91 on a powered-lift shall inspect critical parts in accordance with the maintenance manual or instructions for continued airworthiness (ICA), or as otherwise approved by the President. "Critical part" under the GACAR Part 100 has the same meaning as provided in GACAR § 27.602 and GACAR § 29.602.

2.3 Powered-Lift Operations in Remote Areas Under Part 135. GACAR § 100.402(a) requires CHs operating powered-lift in remote areas under GACAR Part 135 to comply with the preventive maintenance protocols prescribed in GACAR § 43.5(h).

2.4 Inspection of Critical Parts. GACAR §100.402(b) applies the definition of a "critical part" in GACAR §§ 27.602 and 29.602 to powered-lift subject to the inspection outlined in GACAR § 43.23(b), as specified in the aircraft manufacturer's maintenance manual. The intent for critical part inspections will be the same as currently written in GACAR § 43.23 for rotorcraft but may have different nomenclature for powered-lift critical parts as specified in the maintenance manual or ICAs. In lieu of complying with GACAR § 43.23(b), each person performing an inspection required by GACAR Part 91 on a powered-lift shall inspect critical parts in accordance with the maintenance manual or ICA, or as otherwise approved by the President.

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CHAPTER 3. PART 91 OPERATIONS

3.1 Operations of Powered-Lift. Persons operating powered-lift must continue to comply with rules applicable to all aircraft in GACAR Part 91. Table 1 to GACAR § 100.302 outlines the airplane, helicopter, and rotorcraft provisions that are applicable to powered-lift. Moreover, any provisions in GACAR Part 91 that are specific to a category of aircraft such as airplane or helicopters, and that are not outlined in Table 1 to GACAR § 100.302, do not apply to powered-lift.

3.2 GACAR Part 91 Subpart A, General. Powered-lift must comply with the provisions in GACAR Part 91 Subpart A that are applicable to aircraft. Paragraphs 3.2.1 and 3.2.2 below detail how the regulations in GACAR Part 91 Subpart A that contain airplane- or helicopter-specific provisions apply to powered-lift. The GACA also amended a small number of regulations to accommodate powered-lift, discussed herein.

3.2.1 *GACAR § 91.1, Applicability.* The provisions in GACAR Part 91 Subpart A are applicable to all aircraft operating in KSA airspace unless specifically excepted, such as for aircraft governed by GACAR Part 103 or 107. The GACA intends for powered-lift to comply with the provisions in GACAR Part 91 Subpart A.

3.2.2 Reserved.

3.3 Part 91 Subpart B, Flight Rules. Powered-lift must comply with the provisions in GACAR Part 91 Subpart B that are applicable to aircraft. Paragraphs 3.3.1 through 3.3.14 below detail how the regulations in GACAR Part 91 Subpart B that contain airplane or helicopter specific provisions apply to powered-lift.

3.3.1 *GACAR § 91.143, Preflight Action (refer to GACAR § 100.302(b)).* GACAR § 91.43 contains the requirement for a pilot in command (PIC) to be familiar with all available information concerning a flight, including takeoff and landing distance data as specified in the approved aircraft flight manual. Due to the hybrid capabilities of powered-lift, a PIC must be familiar with takeoff and landing distances for all applicable flight modes. GACAR § 100.302(b) requires powered-lift with an aircraft flight manual approved through the certification process to comply with the provisions in GACAR § 91.43.

3.3.2 *GACAR § 91.49, Use of Safety Belts, Shoulder Harnesses, and Child Restraint Systems (refer to GACAR § 100.302(c)).* GACAR § 100.302(c) makes GACAR § 91.49(b) applicable to all powered-lift. In GACAR § 91.49(b) for seaplane and float-equipped rotorcraft operations during movement on the surface, persons pushing off a seaplane or float-equipped rotorcraft from the dock or persons mooring a seaplane or rotorcraft at a dock are excepted from the seating and safety belt requirements. The exceptions also apply to powered-lift in GACAR § 100.302(c). For further information about how to comply with GACAR § 100.302(c), powered-

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lift operators should apply the information contained in the current edition of FAA AC 91-69, Seaplane Safety for Part 91 Operators.

3.3.3 *GACAR § 91.57, Flight Instruction; Simulated Instrument Flight and Certain Flight Tests.* GACAR § 91.57 requires the use of fully functional dual controls for aircraft used in flight instruction with exceptions for free manned balloon and instrument instruction in airplanes with a single functioning throwover control wheel. However, the powered-lift final rule adopted provisions to facilitate pilot certification in powered-lift with a single functioning flight control and single pilot station.

Note: Further discussion of powered-lift and fully functioning dual controls for flight instruction is contained in GACA AC 190-2, Pilot Training and Certification for Powered-Lift Operations.

3.3.4 *GACAR § 91.61, Right-of-Way Rules: Except Water Operations.*

3.3.4.1 GACAR § 91.61 requires all powered-lift to comply with GACAR § 91.61(d)(2) and (d)(3). Under the rule in GACAR § 91.61, when aircraft of the same category are converging at approximately the same altitude (except head-on, or nearly so), the aircraft to the other's right has the right-of-way. When the aircraft are of different categories, GACAR § 91.61(d)(1) through (d)(3) establishes a hierarchy giving priority to balloons, then gliders, followed by airships, and then to airplanes and rotorcraft. An aircraft that is towing or refueling other aircraft has right-of-way over all other engine-driven aircraft. Under the revised GACAR § 91.61, powered-lift, airplanes, and rotorcraft are grouped in the same right-of-way category. Thus, if a powered-lift is converging with an airplane, the aircraft to the right has the right-of-way.

3.3.4.2 To comply with this rule, powered-lift should refer to the current editions of FAA AC 90-48, Pilots' Role in Collision Avoidance, and FAA AC 90-66, Non-Towered Aerodrome Flight Operations. When referring to FAA AC 90-66, powered-lift operators should note that the term "powered-lift" should be considered equivalent to the terms "airplane" or "rotorcraft." For example: A glider has the right-of-way over an airship, powered parachute, weight-shift-control aircraft, airplane, rotorcraft, or powered-lift. Also, an airship has the right-of-way over a powered parachute, weight-shift-control aircraft, airplane, rotorcraft, or powered-lift.

3.3.5 *GACAR § 91.67, Minimum Safe Altitudes: General.* GACAR § 91.67 prescribes the minimum safe altitude (MSA) for aircraft operations. GACAR § 91.67(e) contains an exception for rotorcraft, allowing them to operate below the minimum altitudes prescribed in GACAR § 91.67(b) and (c) when certain conditions are met. Operations below the minimum safe altitudes prescribed in GACAR § 91.67(b) and (c) require a waiver issued under GACAR §§ 91.601 and 91.611 and will never permit operations below 200 feet above ground in vertical-lift flight mode and never below 300 feet above ground when operating in the wing-borne flight mode. Powered-lift operating in the vertical-lift flight mode that have demonstrated the capability to autorotate or conduct an approved equivalent maneuver are allowed the same MSAs as those afforded to helicopters in GACAR § 91.67(e), as outlined in GACAR § 100.302(d). With regard to powered-

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lift utilizing helicopter routes, the GACA notes that some helicopter routes may be lower than a minimum altitude published in the aircraft flight manual for a given powered-lift configuration (e.g., an altitude which enables a transition from wing-borne flight mode to vertical-lift flight mode at an altitude sufficient to conduct a safe autorotation, or an approved equivalent maneuver, to a landing). Regardless of any clearance or helicopter route prescribed altitude, no powered-lift may operate lower than any limitation specified in the aircraft flight manual or any other limitation. The President reiterates that a clearance by air traffic does not grant exemption from any rule. GACAR § 91.13 requires compliance with all aircraft flight manual limitations at all times.

3.3.5A *GACAR § 91.69, Minimum Altitudes for Use of Autopilot.* GACAR § 91.69 details minimum altitudes for use of an autopilot. GACAR § 91.69 is applicable to all aircraft but contains many references to an Airplane Flight Manual in multiple paragraphs and, in GACAR § 91.69(g), excepts rotorcraft operations. GACAR § 100.306(c.1) specifies that GACAR § 91.69(a) through (f) are applicable to powered-lift. It additionally specifies that the requirements referencing the Airplane Flight Manual in GACAR § 91.69(b) are applicable to a Powered-Lift's Aircraft Flight Manual (PLAFM). Other requirements that reference the Airplane Flight Manual in GACAR § 91.69(a) through (f) are also applicable to a PLAFM. GACAR § 91.69(c) stipulates requirements for the use of an autopilot for enroute operations, including climbs and descents. Powered-lift that have a published minimum engagement altitude for enroute operations contained in their aircraft flight manual may use the autopilot for enroute operations at that minimum engagement altitude. If no minimum engagement altitude is specified in the aircraft flight manual, then a powered-lift could not use the autopilot during enroute operations below 500 feet or at an altitude that is no lower than twice the altitude loss specified in the aircraft flight manual for an autopilot malfunction in cruise conditions, whichever is greater. The GACA also retained the provision in GACAR § 91.69(c)(3) that permits the use of the autopilot enroute at an altitude specified by the President, which is applicable to all powered-lift regardless of whether they have an aircraft flight manual minimum engagement altitude or not.

3.3.6 *GACAR § 91.121, Operating On or In the Vicinity of an Aerodrome in Class G Airspace (refer to GACAR § 100.302(e) and (f)).*

3.3.6.1 When entering the traffic pattern to land, powered-lift may transition between wing-borne flight mode and vertical-lift flight mode. Wing-borne flight mode (forward flight) is similar to an airplane, and vertical-lift flight mode (vertical flight) is similar to a helicopter.

3.3.6.2 If the powered-lift intends to land in wing-borne flight mode, GACAR § 100.302(e) requires the pilot to comply with GACAR § 91.121(b)(1) and make all turns of the powered-lift to the left, unless the aerodrome displays approved light signals or visual markings indicating that turns should be made to the right, in which case the pilot must make all turns to the right. If the powered-lift is operating in vertical-lift flight mode, GACAR § 100.302(f) requires the pilot to comply with GACAR § 91.121(b)(2) and avoid the flow of fixed-wing aircraft.

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3.3.6.3 To comply with this rule, powered-lift operators should refer to FAA AC 90-66. Powered-lift pilots must consider their mode of flight in accordance with GACAR § 100.302(e) and (f). For example, FAA AC 90-66, paragraph 12.1.1 states "[i]n the case of a helicopter approaching to land other than on the runway in use, the pilot should avoid the flow of fixed-wing aircraft and land on a marked helipad or suitable clear area." Therefore, powered-lift operating in vertical-lift flight mode would comply with the helicopter provision, avoid the flow of fixed-wing aircraft, and land on a marked helipad or suitable clear area.

3.3.7 *GACAR § 91.127, Operations in Class D Airspace (refer to GACAR § 100.302(h) through (j)).* The provisions of GACAR § 91.127(a) through (d), (g)(1), and (i) refer to aircraft and are applicable to powered-lift.

3.3.7.1 Reserved.

3.3.7.2 *GACAR § 91.127(e)(3) Approaching to Land on a Runway Served by a Visual Approach Slope Indicator (VASI) (refer to GACAR § 100.302(h)).*

3.3.7.2.1 GACAR § 100.302(h) requires a powered-lift pilot intending to land in wing-borne flight mode and approaching a runway served by a VASI to comply with GACAR § 91.127(e)(3). The pilot must operate the powered-lift at an altitude at or above the glidepath until a lower altitude is necessary for a safe landing. The requirement for powered-lift to remain at or above the glidepath ensures adequate obstacle clearance is maintained during the approach.

3.3.7.2.2 Powered-lift pilots intending to land in vertical-lift flight mode are anticipated to be flying more slowly than powered-lift pilots landing in wing-borne flight mode. Thus, the slower speed and the ability to maneuver like a helicopter make it unnecessary to comply with GACAR § 100.302(g)(1).

3.3.7.3 *Section 91.127(f)(1) Class D Airspace Landing Approach Procedures (refer to GACAR § 100.302(i)).*

3.3.7.3.1 GACAR § 100.302(i) requires powered-lift pilots intending to land in wing-borne flight mode to comply with the airplane rule in GACAR § 91.127(f)(1). The pilot must circle the aerodrome to the left unless the powered-lift is conducting a circling approach or if the pilot is otherwise directed by air traffic control (ATC).

3.3.7.3.2 GACAR § 100.302(j) requires powered-lift pilots intending to land in vertical-lift flight mode to comply with the helicopter rule in GACAR § 91.127(f)(2) and avoid the flow of fixed-wing traffic unless the powered-lift is conducting a circling approach or if the pilot is otherwise directed by ATC.

Note: In GACAR § 100.302(i)(2) and (j)(2), the requirements of GACAR § 91.127(f) do not apply to powered-lift conducting a circling approach because a circling approach may have specific procedures established or turns may be requested by ATC to ensure safety in the traffic pattern.

3.3.7.4 Reserved.

3.3.8 Reserved.

3.3.9 *GACAR § 91.161(a) and (b), Fuel Requirements for Flight in VFR Conditions (refer to GACAR § 100.302(l)).*

3.3.9.1 Except in accordance with GACAR § 100.302(l), all powered-lift must comply with the airplane fuel requirements prescribed in GACAR § 91.161(a). In accordance with GACAR § 100.302(l), powered-lift with the performance capability (as outlined in the aircraft flight manual) to conduct a landing in the vertical-lift flight mode along the entire route of flight may use the visual flight rules (VFR) fuel requirements outlined in GACAR § 91.161(b). However, if the powered-lift cannot be assured of a safe landing in vertical-lift flight mode along the entire route of flight, then compliance with GACAR § 91.161(a) would be required.

3.3.9.2 Indications that a powered-lift may not be assured of a safe landing in vertical-lift flight mode along the entire route of flight may include a limitation or requirement in the aircraft flight manual that would preclude such a landing. Likewise, the powered-lift may not be capable of transitioning from wing-borne to vertical-lift flight mode quickly enough to comply. This provision will not prevent a powered-lift from operating in wing-borne flight mode. Rather, it will require the powered-lift to have the performance capability (as detailed in the aircraft flight manual) to conduct a landing in vertical-lift flight mode along the entire route of flight in order to take advantage of the rotorcraft VFR fuel requirements.

Note: The term "fuel" includes any form of energy used by an engine or powerplant installation, such as provided by carbon-based fuels or electrical potential. The term "fuel systems" includes a means of storage for the electrical energy provided (e.g., batteries that provide energy to an electric motor) or devices that generate energy for propulsion (e.g., solar panels or fuel cells).

3.3.10 *GACAR § 91.165, Basic VFR Weather Minimums (refer to GACAR § 100.302(m), (n), and (o)).*

GACAR § 91.165 permits helicopters to operate under lower visibility and cloud clearance minimums than airplanes in Class G Airspace. GACA recognizes that helicopters operate at lower speeds and are capable of greater maneuverability than other aircraft. This allows pilots to see and avoid other air traffic or obstructions in time to prevent a collision. Powered-lift may have equivalent maneuvering capabilities to helicopters when operating in vertical-lift flight mode and therefore may comply with the helicopter provisions in certain circumstances. To comply with these provisions, powered-lift must be operated at a speed that allows the pilot enough time to see and avoid any other air traffic, or any obstructions in time to avoid a collision. A powered-lift that cannot be operated at a speed that allows the pilot enough time to see and avoid any other air traffic or any obstructions to avoid a collision, regardless of the mode of flight, must comply with the airplane visibility minimums prescribed in GACAR § 91.165(b) and (c)(2).

3.3.11 *GACAR § 91.167, Special VFR Weather Minimums (refer to GACAR § 100.302(p)).* GACAR § 91.167 permits helicopters to operate under lower visibility and cloud clearance minimums than airplanes. For the same reasons stated in paragraph 3.3.10 above, powered-lift may comply with the helicopter exceptions of GACAR § 91.167(b)(3), (b)(4), and (c), provided powered-lift are operated at a speed that allows the pilot enough time to see and avoid any other air traffic or any obstructions in time to avoid a collision. As such, GACAR § 100.302(p) stipulates that the helicopter exceptions outlined in GACAR § 91.167(b)(3), (b)(4), and (c) apply to powered-lift operating in vertical-lift flight mode when those aircraft are operated at a speed that allows the pilot to see any other traffic or obstructions in time to avoid a collision.

3.3.12 *GACAR § 91.181, Fuel Requirements for Flight in IFR Conditions (refer to GACAR § 100.302(q)).*

3.3.12.1 GACAR § 91.181(a)(1) through (3) specifies no person may operate a civil aircraft in instrument flight rules (IFR) conditions unless it carries enough fuel (considering weather reports and forecasts and weather conditions) to:

1. Complete the flight to the first aerodrome of intended landing;

2. If necessary, and except as provided in GACAR § 91.181(b), to fly from that aerodrome to the alternate aerodrome; and

3. To fly after that for 45 minutes at normal cruising speed or, for helicopters, fly after that for 30 minutes at normal cruising speed.

3.3.12.2 Because a powered-lift can operate similarly to a helicopter, powered-lift are required to only carry the fuel reserves of a helicopter in certain circumstances. As such, powered-lift authorized to conduct helicopter instrument flight procedures and that have the performance capability, as provided in the aircraft flight manual, for the entire flight to conduct a landing in the vertical-lift flight mode may comply with the IFR fuel requirements established for helicopters.

3.3.12.3 Furthermore, during aircraft certification, the GACA will assess the aircraft's stability, system, and equipage for IFR operations. A powered-lift that does not possess these characteristics may still be certificated for IFR but will be prohibited from performing helicopter instrument flight procedures and have a limitation in the aircraft flight manual to that effect. Powered-lift with such a limitation would not be authorized to use this option and would be required to use the IFR alternate aerodrome minimums specified for airplanes.

3.3.12.4 GACAR § 100.302(q) stipulates that:

1. A powered-lift authorized to conduct helicopter instrument flight procedures, and that has the performance capability for the entire flight to conduct a landing in the vertical-lift flight mode as outlined in the aircraft flight manual, may use the 30-minute fuel requirement established for helicopters under GACAR § 91.181(a)(3);

2. A powered-lift authorized to conduct helicopter instrument flight procedures, and that has the performance capability for the entire flight to conduct a landing in the vertical-lift flight mode as outlined in the aircraft flight manual, may use the helicopter provision under GACAR § 91.181(b)(2)(ii); and

3. A powered-lift that is unable to meet the requirements outlined in GACAR § 100.302(q)(1) and (2) must use the 45-minute fuel requirement outlined in GACAR § 91.181(a)(3) and the aircraft requirement outlined in GACAR § 91.181(b)(2)(i).

3.3.12.5 When taking into consideration the performance capability to conduct a landing in the vertical-lift flight mode, a person must consider the energy required to successfully complete a descent from the altitude they plan to use, any required instrument or visual procedure, and land in the vertical-lift flight mode. There may be performance requirements or limitations contained in the aircraft flight manual, or in any approved minimum equipment list (MEL) that would prevent a powered-lift from conducting a landing in the vertical-lift flight mode, such as a landing weight limitation or a deferred maintenance item, thereby requiring a person to use the 45 minutes of reserve fuel.

3.3.13 *GACAR § 91.185, IFR Flight Plan: Information Required (refer to GACAR § 100.302(r)).*

3.3.13.1 GACAR § 91.185(b)(2)(ii) stipulates that GACAR § 91.185(a)(2) does not apply if appropriate weather reports or weather forecasts, or a combination of them, indicate, for helicopters, at the estimated time of arrival (ETA) and for 1 hour after the ETA, the ceiling will be at least 1,000 feet above the aerodrome elevation, or at least 400 feet above the lowest applicable approach minimums, whichever is higher, and the visibility will be at least 2 statute miles (sm).

3.3.13.2 Likewise, GACAR § 91.185(c)(1)(ii) stipulates that unless otherwise authorized by the President, no person may include an alternate aerodrome in an IFR flight plan unless appropriate weather reports or weather forecasts, or a combination of them, indicate that, at the ETA at the alternate aerodrome, the ceiling and visibility at that aerodrome will be at or above the following weather minimums:

3.3.13.2.1 If an instrument approach procedure (IAP) has been published the minimums for that aerodrome will be:

1. Per GACAR § 91.185(c)(1)(i), for powered-lift operating in the wing-borne flight mode and/or with a flight manual limitation that prohibits helicopter instrument flight procedures, the alternate aerodrome minimums specified in that procedure, or if none are specified, the following approach minimums:

a. For a precision approach procedure, a ceiling of 600 feet and visibility of 2 sm.

b. For a non-precision approach procedure, a ceiling of 800 feet and visibility of 2 sm.

2. Per GACAR § 91.185(c)(1)(ii), for powered-lift operating in the vertical-lift mode, a ceiling of 200 feet above the minimum for the approach to be flown, and visibility of at least 1 sm but never less than the minimum visibility for the approach to be flown.

3.3.13.3 Because powered-lift can operate similarly to helicopters, powered-lift may utilize the requirements for helicopter IFR flight plan required information in certain circumstances. As such, GACAR § 100.302(r) stipulates:

1. Powered-lift authorized to conduct helicopter instrument flight procedures and that have the performance capability to land in the vertical-lift flight mode, as outlined in the aircraft flight manual, may use the helicopter provisions specified in GACAR § 91.185(b)(2)(ii) and (c)(1)(ii); and

2. Powered-lift that are unable to meet the requirements outlined in GACAR § 100.302(r)(1) must use the requirements for aircraft other than helicopters under GACAR § 91.185(b)(2)(i) and (c)(1)(i).

3.3.14 *GACAR § 91.191, Takeoff and Landing Under IFR (refer to GACAR § 100.302(s) and (t)).*

3.3.14.1 GACAR § 91.191(k) provides civil aerodrome takeoff GACAR § 91.175(k)(1) provides takeoff visibility minimums if the takeoff visibility minimums are not otherwise provided under GACAR Part 97 for a particular aerodrome. In this case, a powered-lift with two or less engines must have a minimum visibility of 1600 m for takeoff. For more than two engines, a minimum visibility of 800 m is required for takeoff.

3.3.14.2 GACAR §100.302(s) stipulates that a powered-lift with two engines or less that conducts their takeoff vertically and are authorized to use helicopter instrument flight procedures may use the helicopter minimums of 800 m as provided in GACAR § 91.191(k)(1)(iii).

3.3.14.3 Reserved.

3.3.14.4 Reserved.

3.4 GACAR Part 91 Subpart C, Equipment, Instrument, and Certificate Requirements, and Part 91 Subpart D, Special Flight Operations.

Powered-lift must comply with the provisions in GACAR Part 91 Subpart C and Subpart D that are applicable to aircraft. Paragraphs 3.4.1 through 3.4.7 below detail how the regulations in GACAR Part 91 subparts C and D that contain airplane- or helicopter-specific provisions apply to powered-lift.

3.4.1 *GACAR § 91.303, Instruments and Equipment Requirements: Powered Aircraft with Standard Airworthiness Certificates and Certificate of Authorization under GACAR Part 91 (refer to GACAR § 100.302(u) and (v)).*

3.4.1.1 GACAR § 100.302(u) requires all small powered-lift to comply with GACAR § 91.303(c)(14). GACAR § 91.303(c)(14) requires airplanes to be equipped with a position and anticollision light system that meets the requirements set forth in GACAR § 23.2530(b). Under GACAR § 100.302(u), powered-lift must be equipped with position and anticollision lights that meet the requirements set forth in GACAR § 23.2530(b). In the event of failure of any light of the anticollision light system, operation of the aircraft may continue to a location where repairs or replacement can be made.

3.4.1.2 Reserved.

3.4.1.3 GACAR § 100.302(v) requires powered-lift certified for IFR operations to comply with the airplane provisions in GACAR § 91.303(e)(6). Under GACAR § 100.302(v), powered-lift must be equipped with either a gyroscopic rate-of-turn indicator or a third attitude instrument system usable through flight attitudes of 360 degrees of pitch and roll installed.

3.4.2 *GACAR § 91.303(c)(17), Emergency Locator Transmitters (refer to GACAR § 100.302(w)).*

GACAR § 100.302(w) requires all powered-lift to comply with GACAR § 91.303(c)(17) and be equipped with an emergency locator transmitter (ELT). To comply with this rule, powered-lift operators should refer to the current edition of FAA AC 91-44, Installation and Inspection Procedures for Emergency Locator Transmitters and Receivers. FAA AC 91-44 describes installation and inspection procedures for ELT systems and is intended to be used in conjunction with or as a supplement to the installation, maintenance, and inspection requirements found in the maintenance manuals and other information related to the installation, usage, inspection, and maintenance of ELTs.

3.4.3 *GACAR § 91.309, Inoperative Instruments and Equipment.* GACAR § 91.309 prescribes the circumstances in which an aircraft may be operated with inoperative instruments and equipment. Powered-lift operators must have an approved minimum equipment list (MEL) as required by GACAR § 91.309(a) through (c) in order to take off an aircraft with inoperative instruments or equipment installed.

Note: GACAR § 91.309(d) allows an aircraft without an approved MEL to continue to be operated with inoperative instruments and equipment when certain conditions are met. GACAR § 91.309(d) does not apply to powered-lift and is not included in GACAR Part 100. Powered-lift without an approved MEL must comply with GACAR § 91.309(a) through (c) and may not take off with inoperative instruments and equipment installed.

3.4.4 Reserved.

3.4.5 Reserved.

3.4.6 *GACAR § 91.303(q), Terrain Awareness and Warning System (refer to GACAR § 100.302(y)).*

3.4.6.1 GACAR § 100.302(y) requires powered-lift to comply with GACAR § 91.303 (q) except instead of a Class B terrain awareness and warning system (TAWS) meeting TSO-C151, powered-lift must be equipped with a helicopter terrain awareness and warning system (HTAWS) that meets the requirements in TSO-C194 and Section 2 of RTCA DO-309 (incorporated by reference, see GACAR § 100.9) or an approved TAWS A/HTAWS hybrid system. Powered-lift operators may meet the requirement by having either a traditional HTAWS or a hybrid TAWS A/HTAWS that has been approved by the FAA. The aircraft flight manual must also contain appropriate procedures for the use of the HTAWS and proper flightcrew reaction in response to the HTAWS audio and visual warnings.

3.4.6.2 *HTAWS and Hybrid TAWS A/HTAWS Approvals.*

3.4.6.2.1 Reserved.

3.4.6.2.2 HTAWS that meet the TSO-C194 standard must be approved as part of the type certificate (TC) approval process under GACAR § 21.41(b).

3.4.6.2.3 HTAWS or hybrid TAWS A/HTAWS equipment that does not meet the TSO-C194 standard must be assessed and approved by the GACA prior to its installation and use.

3.4.7 Reserved.

3.5 GACAR Part 91 Subpart E, Maintenance, Preventive Maintenance, and Alterations.

Powered-lift must comply with the provisions in GACAR Part 91 Subpart E that are applicable to aircraft. Paragraphs 3.5.1 and 3.5.2 below detail how the regulations in GACAR Part 91 Subpart E that contain airplane- or helicopter-specific provisions apply to powered-lift.

3.5.1 *GACAR § 91.449, Inspections (refer to GACAR § 100.302(aa)).*

3.5.1.1 GACAR § 100.302(aa) requires technically advanced powered-lift (TAPL) to comply with GACAR § 91.449(e) through (h). TAPL are powered-lift equipped with an electronically advanced system in which the pilot interfaces with a multi-computer system with increasing levels of automation in order to aviate, navigate, or communicate. GACAR § 91.449(e) through (h) sets forth strict and specific inspection requirements for complex aircraft systems to ensure the airworthiness of the aircraft. Powered-lift that are not considered technically advanced must continue to comply with GACAR § 91.449(a), (b), and (d) because those provisions apply to all "aircraft."

3.5.1.2 To be considered a TAPL for purposes of GACAR Part 100 as described in GACAR § 100.302(aa)(1), the powered-lift must be equipped with an electronically advanced multi-computer system that includes one or more of the following installed components:

1. An electronic primary flight display (PFD) that includes, at a minimum, an airspeed indicator, turn coordinator, attitude indicator, heading indicator, altimeter, and vertical speed indicator (VSI);

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2. An electronic multifunction display (MFD) that includes, at a minimum, a moving map using Global Positioning System (GPS) navigation with the aircraft position displayed;

3. A multi-axis autopilot integrated with the navigation and heading guidance system; and

4. An advanced fly-by-wire-flight control system that utilizes electronically operated controls with no direct mechanical link from the pilot to the control surfaces.

3.5.1.3 The display elements described in paragraph 3.5.1.2, items 1 and 2 above must be continuously visible to ensure that the essential aircraft information is displayed and available to the pilot during all phases of flight.

3.5.1.4 The PFD is a display that provides increased situational awareness to the pilot by replacing the traditional six instruments used for instrument flight with an easy-to-scan single display that provides the horizon, airspeed, altitude, vertical speed, trend, trim, and rate-of-turn, among other key relevant indicators. In addition, the PFD is designed specific to controlling the TAPL attitude and altitude relative to the horizon and the surface of the earth, especially when outside visibility is poor or unavailable.

3.5.1.5 The MFD is a display that provides information to the pilot in numerous configurable methods. Often, an MFD will be used in concert with a PFD. The MFD has a different priority; its function is secondary to the PFD. The MFD will have an integrated multi-axis autopilot, navigation, and position awareness information, even though it may include some PFD features for redundancy.

3.5.1.6 Determining whether a display is considered a PFD or an MFD will be based on certain minimum display elements specified by the certifying authority. Using specific characteristics to define TAPL allows the GACA to distinguish between highly complex and less complex powered-lift and to determine which inspection program is required. The GACA and the operator/applicant will determine whether the powered-lift meets the requirements of a TAPL and document the status of the display in the operator's inspection program documents.

3.5.1.7 The rule determines whether a powered-lift constitutes a TAPL because it outlines the required criteria. As part of the operator's certification process, the GACA will communicate with the operator for any specific data (e.g., instructions for continued airworthiness (ICA) and aircraft manufacturer's maintenance manuals) to substantiate the applicant's inspection program.

3.5.1.8 FAA AC 91-90, Part 91 Approved Inspection Programs, and FAA AC 135-10, Approved Aircraft Inspection Program, prescribe the procedures to develop and submit a powered-lift or a TAPL inspection program in accordance with an inspection program selected under GACAR §§ 91.449(f)(4) and 135.245 for review and approval by the GACA. In addition, as explained in FAA AC 91-90, to distinguish between GACAR Part 91 and Part 135 inspection programs, the FAA and GACA generally refers to an inspection program approved under GACAR Part 91 as an approved inspection program (AIP) and refers to an inspection program approved under GACAR

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§ 135.245 as an Approved Aircraft Inspection Program (AAIP). Therefore, the term “AIP” should be used when referring to a program approved under GACAR § 91.449(f)(4), and the term “AAIP” should be used when referring to an inspection program approved under GACAR § 135.245. Note that, as explained in FAA AC 91-90, under GACAR Part 135, operators of aircraft certificated for a passenger seating configuration (excluding any pilot seat) of nine seats or less are not required to have an AAIP but may elect to use one of the inspection programs under GACAR § 91.449, depending on their operation and aircraft fleet size.

3.5.2 *GACAR § 91.451, Altimeter System and Altitude Reporting Equipment Tests and Inspections (refer to GACAR § 100.302(bb)).* GACAR § 100.302(bb) requires all powered-lift to comply with GACAR § 91.451. GACAR § 91.451 prescribes altimeter system and altitude reporting equipment tests and inspections. To comply with this rule, powered-lift operators should refer to the current edition of FAA AC 43-6, Altitude Reporting Equipment and Transponder System Maintenance and Inspection Practices. FAA AC 43-6 provides acceptable methods for testing altimeters, static systems, altitude encoders, and ATC transponder systems.

3.6 Reserved.

3.7 GACAR § 91.411 on REMS and GACAR § 91.416 on Air Ambulance (refer to GACAR Part 91, Subpart D, Section VI and GACAR § 100.302(z.1), (z.4), and (cc).

3.7.1 (1) Subject to the additional requirements prescribed in paragraph (2) below, all the requirements for REMS prescribed in GACAR Part 91, Appendix D, Section VI apply to REMS operations utilizing powered-lift aircraft.

(2) Operating Minimums: During day operations in wing-borne flight mode the weather minimums are prescribed in GACAR § 100.302(cc).

Note:

The President may prescribe additional requirements for REMS operations that utilize powered-lift aircraft that are deemed necessary to maintain an acceptable level of safety for the operation.

3.8 Reserved.

3.9 Other Part 91 Considerations.

3.9.1 *Waivers and Exemptions.* Powered-lift operators that are unable to comply with the requirements in GACAR Part 100 may apply for a waiver from the regulations listed in GACAR § 91.611 or as modified by GACAR Part 100 in accordance with waiver provisions in GACAR § 91.601. If the regulation is not subject to waiver, the operator may seek an exemption in accordance with GACAR Part 11.

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CHAPTER 4. INSTRUMENT FLIGHT PROCEDURES FOR POWERED-LIFT

4.1 Applicability of helicopter instrument flight procedures Under GACAR Part 97 to Powered-Lift. Under GACAR § 100.305, persons operating powered-lift may use helicopter instrument flight procedures if the aircraft is certified for instrument flight rules (IFR) operations and does not contain a limitation prohibiting use of such procedures in its aircraft flight manual.

4.2 Part 97, Standard Instrument Procedures. GACAR Part 97 prescribes instrument flight procedures (IFP) and weather minimums that apply to IFR takeoffs and landings at civil aerodromes in the KSA. Helicopter instrument procedures provide an instrument procedure along a predetermined course to safely allow helicopters to transition from visual meteorological conditions (VMC) to instrument meteorological conditions (IMC) for departures and from IMC to VMC for approaches. The criteria for helicopter instrument flight procedure (approaches or departures) are defined in the current edition of ICAO PANS-OPS and presume a certain level of vehicle performance and stability.

4.3 Instrument Approach Procedures (IAP). Helicopter IAPs are designed presuming nominal descent rates and gradients over a range of given airspeeds. Helicopter instrument approaches also presume the maximum and minimum descent glideslope and gradient that may be encountered while maintaining vertical navigation accuracy. In addition, the design of the IAPs allows for the aircraft to descend to the minimum descent altitude (MDA) or decision altitude (DA) prior to or upon arriving at the missed approach point (MAP). At the MAP, the powered-lift pilot assesses whether the flight can safely and legally proceed to the destination in the meteorological conditions present. Continuation of the flight beyond the MAP must be accomplished via a visual transition segment in accordance with the design of the IAP as prescribed in GACAR § 91.191(c). The MAP is located such that the aircraft can execute the missed approach procedure or visually transition to a safe landing by using a nominal deceleration rate. Both the missed approach procedures and departure procedures are designed with the underlying minimum assumption of aircraft performance as defined in the PANS-OPS criteria.

4.4 Powered-Lift Seeking IFR and Helicopter Instrument Flight Procedure Approval. The type certifying authority will assess powered-lift operators seeking to fly IFR and use helicopter instrument flight procedures. The type certifying authority will consider the aircraft's stability, system, and equipage for IFR operations as compared to helicopters. This assessment will occur during the type certification process as set forth in GACAR § 21.41(b). A powered-lift design that meets these standards can be certificated for IFR flight and authorized to

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conduct helicopter instrument flight procedures. A powered-lift that does not possess these characteristics may still be certificated for IFR but have a limitation from conducting helicopter instrument flight procedures listed in the aircraft flight manual to that effect. Powered-lift operators should review the performance and limitations section of their aircraft flight manual prior to utilizing helicopter instrument flight procedures (IFP).

CHAPTER 5. PART 135 OPERATIONS

5.1 Organization of Topics. This chapter is organized to promote a logical sequence of topics for the reader to easily navigate. The chapter is broken down into two distinct parts: GACAR Part 135 regulations as they pertain to GACAR Part 100; and the FAA and GACA ACs that are impacted by GACAR Part 100.

5.1.1 Reserved.

5.1.2 Much of the information contained in current FAA and GACA ACs that are relevant to GACAR Part 135 operations can be applied to powered-lift operations without any further explanation. The following FAA ACs, which are addressed in paragraph 5.11 below, contain information the GACA believes would be valuable in assisting operators utilizing powered-lift in applying the requirements of the GACAR Part 100:

- FAA AC 120-27, Aircraft Weight and Balance Control.

5.2 Applicability of GACAR Part 100 to Powered-Lift Part 135 Operations.

5.2.1 GACAR § 100.306 contains the provisions under GACAR Part 135 that are applicable to powered-lift (other than the GACAR Part 135 "aircraft" provisions that already apply to powered-lift). GACAR § 100.306 states that no person may operate a powered-lift under GACAR Part 135 unless that person complies with the regulations listed in Table 1 to GACAR § 100.306 despite their applicability to airplanes, rotorcraft, or helicopters, and requires a person operating powered-lift to comply with the requirements of those sections. Additionally, the provisions contained in the second column of Table 1 of GACAR § 100.306, and any additional requirements specified in the third column of Table 1 of GACAR § 100.306 are also applicable.

5.2.2 The GACA also notes that some GACAR Part 135 regulations refer to airworthiness standards in GACAR Part 23, 25, or 27. When an airworthiness standard is referenced in a particular operating rule, those specific standards listed may or may not be used in their entirety due to some of the designs unique to each particular aircraft. When a particular airworthiness certification standard is referenced, but it is not practical to use that standard in its entirety, the GACA, as part of the type certification process for special class aircraft, could instead apply such airworthiness criteria as the certifying authority and GACA may find provide an equivalent level of safety in accordance with GACAR § 21.41(b).

5.3 Reserved.

5.4 GACAR Part 135, Subpart G Rotorcraft Performance Operating Limitations (refer to GACAR § 100.306(--)). As per GACAR § 100.306(--) the rotorcraft performance operating limitations in GACAR Part 135, Subpart G apply to powered-lift only capable of takeoff and landing in vertical-lift flight mode.

5.4.1 When establishing what performance class the aircraft is operating under it is necessary to know what performance category the powered-lift (VTOL) was certificated to. Powered-lift (VTOL) aircraft type certificated to “increased performance” [FA term] or “category enhanced” [EASA term] performance criteria are capable of operating in performance class 1. Powered-lift (VTOL) aircraft type certificated to “essential performance” [FAA term] or “category basic” [EASA term] performance criteria are NOT capable of operating in performance class 1. FAA AC 21.17-4 and EASA Special Condition SC-VTOL provide additional information on these performance categories. The aircraft flight manual will indicate which performance category(s) the aircraft was type certificated to.

5.5 Reserved.

5.6 Reserved.

5.7 GACAR § 135.658, Fuel and Oil Supply.

5.7.1 GACAR § 135.658(c)(5)(i) and (ii) requires airplanes to have enough fuel supply under VFR, considering wind and forecast weather conditions, to reach the first point of intended landing at normal cruise fuel consumption and then fly after that point for 45 minutes for reciprocating engine aircraft or 30 minutes for turbine engine aircraft.

GACAR § 135.658(c)(5)(iii) states that helicopters must have enough fuel to fly to the first point of intended landing, considering wind and forecast weather conditions, and to fly after that for at least 30 minutes (20 minutes when operating within an area providing continuous and suitable precautionary landing sites) regardless of day or nighttime.

5.7.2 GACAR § 100.306(ss) allows a powered-lift to use the VFR helicopter fuel minimums when they are able to conduct a landing in the vertical-lift flight mode. This provision would not prevent a powered-lift from operating in the wing-borne flight mode but will require the powered-lift to have the performance capability, as detailed in the aircraft flight manual, to conduct a landing in the vertical-lift flight mode along the entire route of flight. When using this provision, a person must consider any landing performance data that enables a landing in the vertical-lift flight mode, including taking into consideration the energy requirements to successfully complete a descent and landing in the vertical-lift flight mode from the altitude they plan to use. There may be performance requirements or limitations contained in the aircraft flight manual that could prevent a powered-lift from conducting a landing in the vertical-lift flight mode, such as a landing weight limitation, thereby requiring the use of the airplane fuel supply requirements of GACAR § 135.685(c).

5.7.3 GACAR § 100.306(rr) and (ss) contain the flexibility to apply for a deviation for a fuel supply other than the airplane or helicopter VFR fuel supply requirements prescribed in GACAR § 135.685. This deviation will be available only to those powered-lift operating over specific routes that have predetermined suitable landing areas available. A suitable landing area under

GACAR § 100.306(rr) and (ss) is an area that provides the operator reasonable capability to land without causing undue hazard to persons or property. These suitable landing areas must be site-specific, designated by the operator, and accepted by the GACA.

This will ensure that any operation conducted with less than the prescriptive VFR fuel supply requirements will be conducted under a controlled environment. The controlled environment will incorporate predetermined suitable landing areas that are known by the pilot in advance and be based upon the specific route being flown. The GACA will evaluate each certificate holder's (CH) deviation request to determine if the proposed operation will maintain an equivalent level of safety as currently provided in the prescriptive rule. The authorization to use this deviation would be granted to the CH via OpSpecs.

5.8 GACAR § 135.667, IFR: Alternate Aerodrome Weather Minimums.

5.8.1 GACAR § 135.667 provides the requirements for alternate aerodrome weather minimums. Per GACAR § 135.667(a), no person operating an aircraft other than rotorcraft may designate an alternate aerodrome unless the weather reports or forecasts indicate the weather conditions will be at or above authorized alternate aerodrome landing minimums for that aerodrome at the estimated time of arrival (ETA).

5.8.2 GACAR § 135.667(b) provides alternate aerodrome weather minimums for rotorcraft.

Powered-lift that are authorized to conduct helicopter instrument flight procedures and can land in the vertical-lift flight mode, as provided in the aircraft flight manual, can use the provisions stipulated in GACAR § 135.667(b).

5.9 Reserved.

5.10 GACAR § 135.671 and GACAR § 91.197, Icing Conditions: Operating Limitations (refer to GACAR § 100.306(ww) and (xx)).

5.10.1 GACAR § 135.671 allows an operator to take off in conditions where frost, ice, or snow may reasonably be expected to adhere to the airplane provided the pilot has completed all the applicable training required by GACAR § 135.401 and one of the conditions in GACAR § 135.671(a) through (c) is met. GACAR § 100.306(ww) allows powered-lift to utilize the same provisions as provided in GACAR § 135.671. However, GACAR § 100.306(ww) has some different requirements than those contained in GACAR § 135.671 for airplanes, and are explained below.

5.10.2 In addition to wings and control surfaces, powered-lift may have other surfaces that are negatively impacted by frost, ice, or snow adhering to those surfaces, such as rotor blades. These other surfaces are considered "critical surfaces," which the manufacturer will identify during type certification and will be outlined in the aircraft flight manual for that aircraft. Any frost, ice, or snow adhering to a critical surface could have an adverse impact on the aircraft's ability to operate safely.

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5.10.3 GACAR § 100.306(ww)(1) requires that the powered-lift critical surfaces, as outlined in the aircraft flight manual for that aircraft, must be determined to be free of frost, ice, or snow. Additionally, GACAR § 100.306(ww)(2) stipulates that powered-lift critical surfaces are determined by the manufacturer.

5.10.4 Further discussion of ground deicing/anti-icing programs can be found in the current editions of FAA AC 120-60, Ground Deicing and Anti-Icing Program, and FAA AC 120-107, Use of Remote On-Ground Ice Detection System.

5.10.5 GACAR § 91.197 includes the regulatory requirements for flight into icing conditions, and it specifies that no pilot may fly under IFR into known or forecast light or moderate icing conditions or under VFR into known light or moderate icing conditions unless certain conditions are met. GACAR § 91.197 requires the "aircraft" to have functioning deicing or anti-icing equipment protecting each rotor blade, propeller, windshield, wing, stabilizing or control surface, and each airspeed, altimeter, rate of climb, or flight attitude instrument system. The requirement applies to all aircraft; accordingly, any powered-lift that intends to fly into the icing conditions specified must have functioning deicing or anti-icing equipment.

5.10.6 Reserved.

5.10.7 GACAR § 100.306(xx) prohibits any pilot from flying a powered-lift under IFR into known or forecast icing conditions or under VFR into known icing conditions unless it has been type certificated and appropriately equipped for operations in icing conditions.

5.11 FAA ACs Requiring Further Explanation for the Integration of Powered-Lift.

To comply with GACAR § 100.306, as well as other regulations not contained in GACAR Part 100 but relevant to powered-lift operations, some information from existing FAA ACs was extracted to provide additional information to assist an operator in applying the requirements of GACAR Part 100. This additional information is listed below.

5.11.1 *FAA AC 120-27, Aircraft Weight and Balance Control.* FAA AC 120-27 was written for both airplanes and helicopters, and the majority of its content applies to powered-lift without the need for additional explanation. However, the information regarding standard average weights warrants additional explanation because it will also apply to powered-lift authorized to conduct air ambulance and/or REMS operations.

5.11.1.1 *Standard Average Weights.* Use of standard average weights is not permitted for powered-lift aircraft operating under GACAR Part 100.

5.11.2 Reserved.

5.11.3 Reserved.

5.11.4 *FAA AC 135-14, Helicopter Air Ambulance Operations.*

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5.11.4.1 *Air Ambulance and REMS Operations.* FAA AC 135-14 provides information on HAA operations and was written to specifically address helicopters. Some of the information contained in FAA AC 135-14 also applies to powered-lift conducting REMS and/or air ambulance operations. The information in FAA AC 135-14 reads in conjunction with the information contained within this AC to assist GACAR Part 135 CHs conducting air ambulance operations in powered-lift in complying with GACAR § 91.411, GACAR § 91.416 and GACAR Part 100.

5.11.4.2 *Terms and Requirements.* FAA AC 135-14 contains the terms "helicopter" and "HAA". These helicopter-specific terms and requirements are noted below:

5.11.4.2.1 When the term "helicopter" is used, the topic being addressed in FAA AC 135-14 will be applicable to powered-lift unless otherwise noted.

5.11.4.2.2 When the term "HAA" is used, it will also apply to operators using powered-lift for air ambulance operations under GACAR § 100.302(z.4).

5.11.4.3 *Definitions and Abbreviations*.

5.11.4.3.1 *Autorotational Distance.* Autorotational distance is the distance a powered-lift or rotorcraft can travel in autorotation as described by its manufacturer in the approved aircraft flight manual.

5.11.4.3.2 *Overwater Flight.* FAA AC 135-14 refers to overwater flight for rotorcraft. Overwater flight for powered-lift is the operation of a powered-lift beyond the autorotational or gliding distance from the shoreline. Determining whether the powered-lift is conducting overwater flight depends on the powered-lift's ability to autorotate or glide to reach the shoreline. If the powered-lift is operating over water beyond either the gliding or autorotational distance of the shoreline, then it meets the definition of overwater flight.

5.11.4.3.3 *Shoreline.* Shoreline is land adjacent to the water of an ocean, sea, lake, pond, river, or tidal basin that is above the high-water mark where a powered-lift could be landed safely. This does not include land areas unsuitable for landing, such as vertical cliffs or land intermittently under water.

5.11.4.3.4 *Suitable Offshore Vertiport.* Suitable offshore vertiport is a vertiport that can support the size and weight of the powered-lift being operated where a safe landing can be made. Powered-lift that are able to conduct vertical takeoff and landings can use any suitable vertiport. Additionally, in order to use a vertiport, a powered-lift must be capable of meeting the performance requirements for that specific vertiport, and the powered-lift cannot exceed the size and weight limitations established for that vertiport.

5.11.5 *Certification and HAA-Specific Considerations*. REMS and Air Ambulance operations using helicopters (FAA calls these operations HAA) are authorized through the issuance of OpSpecs. Powered-lift conducting air ambulance operations will experience the same organizational and operational challenges that current HAA operators face. Therefore, the GACA

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will authorize powered-lift operators wanting to conduct REMS and/or air ambulance operations by issuing OpSpecs. In addition, powered-lift air ambulance operators are required by GACAR § 100.302(z.1) and/or (z.4) to comply with any additional requirements imposed by the President to ensure an acceptable level of safety.

5.11.6 *Regulatory Operational Considerations.* HAA operators and persons utilizing powered-lift for air ambulance operations are subject to regulatory operational requirements contained in GACAR 91 and GACAR Part 100.

5.11.7 *Pilot Qualifications*. Refer to GACA AC 100-2, Pilot Training and Certification for Powered-Lift Operations, for detailed information on pilot training and certification and guidance for pilots, examiners, and instructors on the pilot training and certification process in a powered-lift.

5.11.8 Reserved.

5.11.9 *VFR Flight Planning Documentation.*

5.11.9.1 Operators conducting air ambulance operations with powered-lift must comply with GACAR Part 91, including the powered-lift specific requirements contained in § 100.302(z.1) and/or (z.4). GACAR § 100.302(cc) contains different altitude clearance requirements for powered-lift that will be operated en route in the wing-borne flight mode than those currently listed for helicopters. Therefore, GACAR § 100.302(cc) requires the pilot in command (PIC) of a powered-lift that will be operating in the wing-borne flight mode to use the higher minimum altitudes specified for airplanes in GACAR § 91.451 while conducting VFR operations en route. Therefore, to comply with the requirements of GACAR § 100.302(cc), the PIC of a powered-lift being operated in the wing-borne flight mode must ensure that all terrain and obstacles along the route of flight are cleared vertically and horizontally by no less than 500 feet during the day.

5.11.9.2 When a powered-lift will be operated en route in the vertical-lift flight mode the altitude clearance requirements of GACAR § 91.451, those stipulated for helicopters, can be used. The PIC must ensure that all terrain and obstacles along the route of flight are cleared vertically by no less than 300 feet for day operations and 500 feet for night operations.

5.11.10 *Use of Minimum Safe Cruise Altitudes.* GACAR Part 100 requires that prior to conducting VFR operations, the PIC of a powered-lift, when conducting their preflight analysis, must use the minimum safe cruise altitudes specified in GACAR § 100.302 (cc) as explained in paragraph 5.11.9 above.

5.11.11 Reserved.

5.11.12 *Hazards to Operations: Identification and Mitigation.*

5.11.12.1 *Flight Controls.* Leaving the flight controls of a powered-lift while rotors or powerplants are running is a potentially hazardous situation that may be encountered in air ambulance operations. While current regulations do not prohibit the pilot from leaving the controls while the powered-lift is operating, air ambulance operators are encouraged to include procedures for accomplishing this safely in their documented operational procedures and training.

5.11.12.2 *Vertiports/Landing Zones (LZ).* When GACAR Part 135 HAA operations are conducted from established vertiports, those vertiports should meet the criteria established in the current edition of GACA AC on Vertiports, to the maximum extent possible. Powered-lift that are able to conduct vertical takeoff and landings can use any suitable vertiport or vertiport. Additionally, in order to use a vertiport, a powered-lift must be capable of meeting the performance requirements for that specific vertiport, and the powered-lift cannot exceed the size and weight limitations established for that vertiport.

5.11.13 *Operations Under Special Conditions.*

5.11.13.1 *Flat Light, Whiteout, and Brownout.* In accordance with GACAR § 135.349(a)(9), all powered-lift pilots must be tested on procedures for aircraft handling in flat-light, whiteout, and brownout conditions, including methods for recognizing and avoiding those conditions. Air ambulance operators are susceptible to all of these conditions due to the nature of off-aerodrome landings and operating in remote environments. The following descriptions are not intended to be scientific explanations but serve as operational definitions suitable for use by air ambulance operators. These terms should not be used interchangeably.

5.11.13.2 *Flat Light.* Flat light is an optical condition, also known as sector or partial whiteout. It is not as severe as whiteout, but this condition causes pilots to lose depth-of-field and vertical orientation. Flat-light conditions are usually the result of overcast skies over snow or ice fields, inhibiting visual reference. Such conditions can occur anywhere in the world, primarily in snow-covered areas but they can also occur in dust, sand, mud flats, or on glassy water. Flat light can completely obscure features of the terrain, creating an inability to distinguish distances and closure rates. As a result of this reflected light, it can give pilots the illusion of ascending or descending when actually flying level. However, with good judgment and proper training and planning, it is possible to safely operate aircraft in flat-light conditions.

5.11.13.3 *Whiteout/Brownout.* This effect typically occurs when a powered-lift takes off or lands on a dusty or snow-covered area. The rotor or vertical-lift device downwash picks up particles and recirculates them. The effect can vary in intensity depending upon the amount of light on the surface. This phenomenon can happen on the sunniest, brightest day with good contrast everywhere. However, when it happens, there can be a complete loss of visual cues. If the pilot has not prepared for this immediate loss of visibility, the results can be disastrous.

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CHAPTER 6. RESERVED

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APPENDIX A. LIST OF ACRONYMS

Acronym Definition

AAIP Approved Aircraft Inspection Program

AC Advisory Circular

AIP Approved Inspection Program

ATC Air Traffic Control

CG Center of Gravity

CH Certificate Holder

DA Decision Altitude

DRS Dynamic Regulatory System

ELA Electrical Load Analysis

ELT Emergency Locator Transmitter

ETA Estimated Time of Arrival

FAA Federal Aviation Administration

GPS Global Positioning System

GS Glideslope

HTAWS Helicopter Terrain Awareness and Warning System

IAP Instrument Approach Procedure

ICA Instructions for Continued Airworthiness

ICAO International Civil Aviation Organization

IDF Initial Departure Fix

IFP Instrument Flight Procedure

IFR Instrument Flight Rules

IMC Instrument Meteorological Conditions

LZ Landing Zone

MAP Missed Approach Point

MDA Minimum Descent Altitude

MEL Minimum Equipment List

MFD Multifunction Display

MHz Megahertz

MSA Minimum Safe Altitude

NM Nautical Mile

OEM Original Equipment Manufacturer

OpSpec Operations Specification

PFD Primary Flight Display

PIC Pilot in Command

PLAFM Powered-Lift's Aircraft Flight Manual

RA Radio Altimeter

SIC Second in Command

SID Standard Instrument Departure

TAPL Technically Advanced Powered-Lift

TAWS Terrain Awareness and Warning System

TAWS A Class A TAWS

TC Type Certificate

TSO Technical Standard Order

VASI Visual Approach Slope Indicator

VFR Visual Flight Rules

VMC Visual Meteorological Conditions

VSI Vertical Speed Indicator

VTOL Vertical Takeoff and Landing

APPENDIX B. RELATED READING MATERIAL

B.1 The FAA publishes many ACs that address a wide range of topics and audiences. Some of these ACs are applicable to certain kinds of aircraft, while others are not aircraft-specific and do not include aircraft in the subject matter. These FAA ACs may be relevant to powered-lift, even though the term “powered-lift” is not found within each AC. The GACA encourages persons to review the available GACA and FAA ACs to determine the information that may be relevant to their intended operations. All FAA ACs can be found on the FAA website at https://www.faa.gov/regulations_policies/advisory_circulars.

B.2 FAA ACs that are applicable to all aircraft are applicable to powered-lift and other VTOL capable aircraft.

B.3 The list below contains FAA ACs that the FAA and GACA have determined could apply to powered-lift as written with no further explanation required. The GACA acknowledges that the list is not exhaustive but is included here as a guide to assist the reader in identifying potentially relevant ACs.

B.3.1 *General:*

- FAA AC 00-63, Use of Flight Deck Displays of Digital Weather and Aeronautical Information.

Note: Also refer to FAA-H-8083-28, Aviation Weather Handbook.

B.3.2 *Maintenance.*

- FAA AC 43.13-2, Acceptable Methods, Techniques, and Practices – Aircraft Alterations.

B.3.3 *Airman.*

- GACA AC 100-02, Pilot Training and Certification for Manned Powered-Lift (VTOL) Operations
- FAA AC 61-134, General Aviation Controlled Flight Into Terrain Awareness.

B.3.4 *Air Traffic and General Operating Rules:*

- FAA AC 90-23, Aircraft Wake Turbulence.
- FAA AC 90-106, Enhanced Flight Vision System Operations.
- FAA AC 90-108, Use of Suitable Area Navigation (RNAV) Systems on Conventional Routes and Procedures.
- FAA AC 91-32, Safety in and Around Helicopters.
- FAA AC 91-79, Aircraft Landing Performance and Runway Excursion Mitigation.
- FAA AC 91-90, Part 91 Approved Inspection Programs.
- FAA AC 135-10, Approved Aircraft Inspection Program.